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<u>Proof</u> (of Schur's Theorem)

Suppose $A x^{(1)} = \lambda_1 x^{(1)}$, $\|x^{(1)}\|_2 = 1$

(otherwise, scale $x^{(1)}$ by $\frac{1}{\|x\|_2}$)

Extend $x^{(1)}$ into a basis

$$\{x^{(1)}, y^{(2)}, \ldots, y^{(n-1)}\}$$

for $\mathbb{C}^n$. Use Gram-Schmidt orthogonalization & then normalize to produce an <u>orthonormal</u> basis

$$\{x^{(1)}, z^{(2)}, \ldots, z^{(n)}\}$$

for $\mathbb{C}^n$. Construct unitary matrix

$$U_1 = \left[x^{(1)} \,|\, z^{(2)} \,|\, \cdots \,|\, z^{(n)} \right] \in M_n.$$

The observe that

$$A U_1 = \left[A x^{(1)} \,|\, A z^{(2)} \,|\, \cdots \,|\, A z^{(n)} \right]$$

$$= \left[\lambda_1 x^{(1)} \,|\, A z^{(2)} \,|\, \cdots \,|\, A z^{(n)} \right]$$

$$\underset{\text{to verify}}{\overset{\text{claim}}{=}} U_1 \left[\begin{array}{c|c} \lambda_1 & * \\ \hline 0 & \\ \vdots & A_1 \\ 0 & \end{array} \right], \quad \text{for some } A_1 \in M_{n-1} \text{ \& } *.$$

Note that $\sigma(A) = \{\lambda_1\} \cup \sigma(A_1)$

i.e. $\sigma(A_1) = \{\lambda_2, \dots, \lambda_n\}$

since $\left[\begin{array}{c|c} \lambda_1 & * \\ \hline 0 & A_1 \end{array}\right]$ is similar to A

Work now with $A_1 \in M_{n-1}$. As above we can find a unitary matrix $U_2 \in M_{n-1}$ such that

$$U_2^* A_1 U_2 = \left[\begin{array}{c|c} \lambda_2 & * \\ \hline 0 & \\ \vdots & A_2 \\ 0 & \end{array}\right], \quad A_2 \in M_{n-2}$$

$$\sigma(A_2) = \{\lambda_3, \lambda_4, \dots, \lambda_n\}$$

Letting

$$V_2 = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & & & \\ \vdots & & U_2 & \\ 0 & & & \end{bmatrix} \in M_n$$

we see that $V_2^* V_2 = I$, i.e. V_2 is unitary.

Also

$$V_2^* U_1^* A U_1 V_2 = \left[\begin{array}{cc|c} \lambda_1 & * & * \\ 0 & \lambda_2 & \\ \hline \vdots & 0 & \\ \vdots & \vdots & A_2 \\ 0 & 0 & \end{array}\right]$$

Continue by working with A_2 etc.

In the same manner we find that there exists a unitary matrix

$$U = U_1 V_2 V_3 \cdots V_{n-1}$$

such that

$$U^* A U = \underbrace{\begin{bmatrix} \lambda_1 & * & * & & * \\ 0 & \lambda_2 & * & & \vdots \\ \vdots & 0 & \lambda_3 & \ddots & * \\ \vdots & \vdots & 0 & \ddots & \\ 0 & 0 & 0 & \cdots\ 0 & \lambda_n \end{bmatrix}}_{=T}$$

where T is now upper triangular and the diagonal entries are the eigenvalues of A (in the order we decided to use them) ∎

<u>Notes</u>

(1) The technique used to work successively with smaller matrices $A, A_c, \ldots$ is known as a "deflation process". Useful in eigenvalue problems (& invariant subspace problems)

(2) Neither U nor T is unique in Schur's Theorem. The choice of how we extend bases & the choice of the order of the eigenvalues lead to different U & T.

(3) It follows from proof of Schur's Theorem that if $AB = BA$, then there exists a <u>common</u> unitary U such that

$$U^* A U, \quad U^* B U$$

are both triangular.

(simultaneous triangularizability)

(The converse does <u>not</u> work: simultaneously triangu/ble $\not\Rightarrow$ commutativity)

§2.5 Some implications of Schur's Theorem

(1) Proof that $\det A = \prod_{\lambda \in \sigma(A)} \lambda$

$$\operatorname{trace}(A) = \sum_{\lambda \in \sigma(A)} \lambda$$

First notice $\forall A, B \in M_n$,

$$\operatorname{trace}(AB) = \operatorname{trace}(BA) \quad (\#)$$

(prove!, easy)

So, let $A = UTU^*$ as in Schur's Theorem

Then

$$\operatorname{trace}(A) = \operatorname{trace}(UTU^*)$$

$$\overset{(\#)}{=} \operatorname{trace}(TU^*U)$$

$$= \operatorname{trace}(T) = \sum_{i=1}^{n} \lambda_i, \ \lambda_i \in \sigma(A)$$

Also

$$\det A = \det(UTU^*) = \cancel{\det U} \cdot \det T \cdot \cancel{\det U^*}$$

$$\overset{U^* = U^{-1}}{=} \det T = \prod_{i=1}^{n} \lambda_i, \ \lambda_i \in \sigma(A).$$

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More implications:

(2) The Cayley-Hamilton Theorem states that

$$P_A(t)\Big|_{t=A} = 0 = \text{zero matrix}$$

$$P_A(A) = 0 \quad , \quad \boxed{P_A(t) = \det(tI - A)}$$

You were asked to prove this when A is diagonalizable (homework)

General proof?

Clarification: $q(t) = a_n t^n + a_{n-1} t^{n-1} + \dots + a_1 t + a_0$

$$= a_n (t - r_1)(t - r_2) \dots (t - r_n)$$

where $q(r_i) = 0$, $i = 1, 2, \dots, n$

By $q(A)$ (evaluating at $t = A$) we mean sub t for A:

$$q(A) = a_n A^n + a_{n-1} A^{n-1} + \dots + a_1 A + a_0 I$$

$$= a_n (A - r_1 I) \dots (A - r_n I) \in M_n$$

So for $P_A(t) = \det(tI - A)$

here plugging $t = A$ does not make sense

NO SENSE: $\underbrace{P_A(A)}_{\text{matrix}} = \det(A \cdot I - A) = \det(0) = \underbrace{0}_{\text{scalar}}$

<u>Theorem</u> (Cayley-Hamilton)

Let $A \in M_n$ and suppose $p_A(t) = \det(tI - A)$ (characteristic polynomial of A).

Then $p_A(A) = 0 \in M_n$.

<u>Proof</u> Suppose

$$p_A(t) = (t-\lambda_1)(t-\lambda_2)\cdots(t-\lambda_n)$$

where $\sigma(A) = \{\lambda_1, \lambda_2, \ldots, \lambda_n\}$.

Let $A = UTU^*$ as in Schur's Theorem.

Then

$$p_A(A) = p_A(UTU^*)$$

$$= (UTU^* - \lambda_1 UU^*)(UTU^* - \lambda_2 UU^*) \cdots (UTU^* - \lambda_n UU^*)$$

$$\overset{U^*U = I}{=} U(T - \lambda_1 I)\cancel{U^* U}(T - \lambda_2 I)\cancel{U^*} \cdots \cancel{U}(T - \lambda_n I)U^*$$

$$= U\underbrace{(T - \lambda_1 I)(T - \lambda_2 I)\cdots(T - \lambda_n I)}U^*$$

This is a product of triangular matrices such that the k-th diagonal entry of the k-th factor equals 0

(because $T = \begin{bmatrix} \lambda_1 & & * \\ & \lambda_2 & \\ 0 & & \ddots \; \lambda_n \end{bmatrix}$)

Exercise! → We can prove (inductively) that such a product is the zero matrix, i.e. $p_A(A) = U \cdot 0 \cdot U^* = 0$ ∎

The inductive hypothesis in the exercise within the proof is:

$$\begin{bmatrix} 0 & t_{12} \\ 0 & t_{22} \end{bmatrix} \begin{bmatrix} t_{11} & t_{12} \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \checkmark$$

Two wrong proof of Cayley-Hamilton

⟶

Two wrong proofs

(i) $P_A(t) = \det(tI - A)$

$$\Rightarrow \quad P_A(A) = \det(A \cdot I - A)$$
$$= \det(0) = 0$$

Does <u>not</u> hold water because it showed matrix = scalar.

WRONG This happened we're not supposed to plug in $t = A$ in "$\det(tI - A$".

(ii) $P_A(t) = \det(tI - A)$, $P_A(\lambda) = 0$, $\forall \lambda \in \sigma(A)$.

We've learned (look at first week notes)

$$\sigma(P_A(A)) = \{P_A(\lambda_1), P_A(\lambda_2), \dots, P_A(\lambda_n)\}$$

where $\sigma(A) = \{\lambda_1, \lambda_2, \dots, \lambda_n\}$

$$= \{0, 0, \dots, 0\}$$

WRONG i.e. "all eigenvalues of $P_A(A)$ are 0 and so $P_A(A)$ is the zero matrix."

Not true: e.g. $A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$, $\sigma(A) = \{0, 0\}$, $A \neq 0$.

<u>Aside:</u> If $A \in M_n$ and $\sigma(A) = \{0, 0, \dots, 0\}$, we call A a <u>nilpotent</u> matrix.

<u>Fact 1</u> $A \in M_n$ is nilpotent <u>iff</u> $\exists k$ such that $A^k = 0$.

e.g. • $\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ ✓

$A \neq 0$, $A^2 = 0$

• $\begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

So $A = \begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix}$ has $\sigma(A) = \{0, 0\}$

<u>Fact 2</u> The smallest k above is always $k \leq n$.

(3) (implication)

Think about the vector space M_n (under addition / scalar multiplication)

Basis? All

$$\left\{ M_{ij} \in M_n : (i,j)^{th} \text{ entry of } M \text{ is one } \& \text{ all other entries are } 0 \right\}$$

$$\left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}$$

is a basis for M_2

So $\dim M_n = n^2$

So given $A \in M_n$, the set

$$\{ I, A, A^2, \ldots, A^{n^2-1} \}$$

is either linearly independent (a basis) or A must be a power A^k that linearly depends on $I, A, \ldots, A^{k-1}$.

We can do a <u>lot</u> better:

<u>FACT</u>: $A \in M_4$, A^n can be written as a linear combination of $I, A, \ldots, A^{4n}$, always?

So $\{I, A, A^2, A^3, \ldots, A^{15}\}$ for $A \in M_4$ is always lin. dependent.

in fact $\{I, A, A^2, A^3, A^4\}$ is lin. dependent.

<u>Prove FACT</u>: (TBC) $\longrightarrow$

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Let $A \in M_n$, $P_A(t) = \det(tI - A)$

$$= t^n + a_{n-1} t^{n-1} + \dots + a_1 t + a_0$$

By Cayley-Hamilton,

$$P_A(A) = 0 \implies A^n + a_{n-1} A^{n-1} + \dots + a_1 A + a_0 I = 0 \quad (*)$$

$$\implies \boxed{A^n = -a_{n-1} A^{n-1} - \dots - a_1 A - a_0 I}$$

As a consequence, <u>all</u> powers of A are in the span of $\{I, A, A^2, \dots, A^{n-1}\}$

In fact, multiplying $(*)$ by A^{-1}, assuming A is invertible, we get

$$A^{n-1} = -a_{n-1} A^{n-2} - \dots - a_1 I - a_0 A^{-1}$$

$$\implies -a_0 A^{-1} = A^{n-1} + a_{n-1} A^{n-2} + \dots + a_1 I$$

$$\overset{a_0 = \det A \neq 0}{\implies} \boxed{A^{-1} = -\frac{1}{a_0} A^{n-1} - \frac{a_{n-1}}{a_0} A^{n-2} \dots - \frac{a_1}{a_0} I}$$

So all integer powers of an invertible matrix belong to span $\{I, A, \dots, A^{m-1}\}$.

<u>Important consequence</u>

Consider $e^z = \sum_{n=0}^{\infty} \frac{z^n}{n!}$

We can consider the <u>matrix exponential</u> function

$$e^A = \sum_{n=0}^{\infty} \frac{A^n}{n!}, \quad A \in M_m \quad m \times m$$

because

$$\|e^A\| = \left\|\sum_{n=0}^{\infty} \frac{A^n}{n!}\right\| \leq \sum_{n=0}^{\infty} \frac{\|A^n\|}{n!}$$

submultiplicative matrix norm

$\downarrow$

later

$$\leq \sum_{n=0}^{\infty} \frac{\|A\|^n}{n!}$$

$$= e^{\|A\|} < \infty$$

i.e. e^A is a convergent series.

How do we compute e^A for a given A?

"Perhaps" we can use this newly discovered consequence of Schur's Theorem (Cayley-Hamilton)

Schur's Theorem can also be used in perturbation results.

<u>Theorem 1</u> Let $A = [a_{ij}] \in M_n$. Then for every $\varepsilon > 0$, there exists $A(\varepsilon) = [a_{ij}(\varepsilon)] \in M_n$ with n distinct eigenvalues and such that

$$\sum_{i,j=1}^{n} |a_{ij} - a_{ij}(\varepsilon)|^2 < \varepsilon$$

(Converse true as well: $\underset{\text{diagonal}}{\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}}$, $\underset{\text{non-diagonalizable}}{\begin{bmatrix} 1 & \varepsilon \\ 0 & 1 \end{bmatrix}}$)

<u>Proof</u> As per Schur's Theorem

$$U^* A U = T, \quad U \text{ unitary}, \ T \text{ triangular}$$

$T = [t_{ij}]$

Choose diagonal matrix

$$E = \begin{bmatrix} e_1 & & & 0 \\ & e_2 & & \\ & & \ddots & \\ 0 & & & e_n \end{bmatrix} \in M_n$$

such that $|e_i| < \sqrt{\frac{\varepsilon}{n}}$ for a given $\varepsilon > 0$.

and $t_{ii} + e_i \neq t_{jj} + e_j, \ i \neq j$

Let $A(\varepsilon) := A + UEU^*$

Notice

$$U^* A(\varepsilon) U = U^* A U + E = T + E$$

which is upper triangular with distinct diagonal entries. So $T+E$ is diagonalizable!

Thus $A(\varepsilon)$ is similar to a diagonalizable matrix $T+E$ and

$$A - A(\varepsilon) = -UEU^*$$

$$\Rightarrow \sum_{i,j=1}^{n} |a_{ij} - a_{ij}(\varepsilon)|^2 = \sum_{i,j=1}^{n} |(-UEU^*)_{ij}|^2$$

$$= \sum_{i,j=1}^{n} |(UEU^*)_{ij}|^2$$

$$= \sum_{i=1}^{n} |e_i|^2 \leq \sum_{i=1}^{n} \left(\sqrt{\frac{\varepsilon}{n}}\right)^2$$

Theorem we proved!

$$= \varepsilon$$

So

Every $A \in M_n$ is arbitrarily close to a matrix with distinct eigenvalues, i.e. diagonalizable.

Matrices with distinct eigenvalues are dense in M_n.